

Script-Controlled ACL – Restrict Record Access Based on Field Value

Naan Mudhalvan / Skill Wallet Project - Testing & QA Phase Document

1. Project & Team Information

- Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value
- Phase: Testing & Quality Assurance Phase
- Team ID: SWTID-2026-9611
- Team Leader: Adhsa Rani P
- Team Members: Gayathri A, Parkavi P, Priyadharshini M, Chinthanika P
- Platform / Environment: ServiceNow / SkillWallet (SmartBridge)

2. Testing Strategy & Objectives

- Objective: Validate that script-controlled Access Control Lists (ACLs) dynamically evaluate user roles and record field values across all CRUD operations on the u_institution_details table.
- Strategy:
 1. Role & Permission Matrix Verification: Test access permissions across custom roles (bb1, bb2, bb3, bb4).
 2. Dynamic Field Condition Evaluation: Test field-level restrictions based on u_branch, state, and is_confidential.
 3. UI & Cross-Interface Consistency: Validate enforcement across Form Views, List Views, and Service Portal interfaces.
 4. Negative & Fallback Testing: Verify zero-trust default-deny behavior for invalid requests or unhandled cases.

3. Test Execution Matrix & Results

Test ID	Operation	Test Role	Field Value Condition	Expected Result	Actual Result	Status
TC-TST-01	DELETE	User with bb4	u_branch == 'EEE'	Delete button visible; deletion succeeds.	Delete button visible; deletion succeeds.	PASSED

Test ID	Operation	Test Role	Field Value Condition	Expected Result	Actual Result	Status
TC-TST-02	DELETE	User with bb4	u_branch == 'CSE'	Delete button hidden; operation blocked.	Delete button hidden; operation blocked.	PASSED
TC-TST-03	DELETE	User with bb3	u_branch == 'EEE'	Access denied due to missing role bb4 .	Access denied due to missing role bb4 .	PASSED
TC-TST-04	WRITE	User with bb3	state == 'Active'	Form editable; updates saved.	Form editable; updates saved.	PASSED
TC-TST-05	WRITE	User with bb3	state == 'Closed'	Form read-only; save blocked.	Form read-only; save blocked.	PASSED
TC-TST-06	READ	User with bb1	is_confidential == true	Record hidden from list view and direct URL access.	Record hidden from list view and direct URL access.	PASSED
TC-TST-07	READ	User with bb3	is_confidential == true	Record visible in list view and form view.	Record visible in list view and form view.	PASSED

4. Interface Consistency & Performance Verification

- Form Views: UI fields and action buttons dynamically toggle visibility based on ACL runtime evaluation.
- List Views: Restricted records are filtered automatically without affecting list navigation or breaking pagination.

- Service Portal: Portal widgets and backend REST APIs enforce server-side ACL security rules consistently.
- Performance Impact: Sub-millisecond execution time per record check; no rendering delay on list queries.

5. Testing Phase Conclusion

All functional, security, and performance test cases passed successfully. The script-controlled ACL implementation is validated for final project delivery under the Naan Mudhalvan initiative.